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Brazilian vehicular emission inventory software – BRAVES

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ABSTRACT

In this work, we present the BRAzilian Vehicular Emissions inventory Software - BRAVES which combines a bottom-up method with fleet information of each county. The new model uses a probabilistic approach that accounts for the fleet characteristics, fuel consumption, vehicle deterioration, and intensity of use, to calculate the vehicular emissions from the exhaust, tires, roads, brakes wear, soil resuspension, refueling, and evaporative emissions. Brazil is the fourth largest vehicle producer in the world. However, there is a lack of detailed information on vehicular emissions, especially in middle and small counties. BRAVES can estimate emissions of a larger variety of pollutants at the county scale and segregate the emissions by formation process and fleet category, a major advantage over national and state inventories. BRAVES is a compromise of more sophisticated models that provide detailed information in data-scarce areas, like small counties and could be easily adapted to other developing countries.

1. Introduction

In this work, we present the BRAzilian Vehicular Emissions inventory Software - BRAVES which combines a bottom-up method with county information. The new model uses a probabilistic approach that accounts for the fleet characteristics, fuel consumption, vehicle deterioration, and intensity of use, to calculate the vehicular emissions from the exhaust, tires, roads, brakes wear, soil resuspension, refueling, and evaporative emissions.

There is a large number of models and databases for developing vehicular inventories, mainly based on top-down (Andreão et al., 2020; Davis et al., 2005; Tuia et al., 2007) or bottom-up (Abou-Senna et al., 2013; Tang et al., 2016; Wang et al., 2009, 2008) approaches. The top-down based methods provide a screening view of vehicular emissions, generally in national and global scales. However, they cannot detail emission characteristics, temporal and spatial variability, and the contribution of each fleet category (Wang et al., 2009). The bottom-up approach used on MOVES (USEPA, 2020a, 2020b, 2020c, 2020d), MOBILE (USEPA, 2002), COPERT (Ntziachristos and Samaras, 2000), and VEIN (Ibarra-Espinosa et al., 2018) are the most resolved and reliable methods. These models require refined input data (ex: acceleration/deceleration and vehicular flow) and computational resources, often not available in most countries (Achour et al., 2011; Goel and Guttikunda, 2015; Goyal et al., 2013; Ibarra-Espinosa et al., 2018; Kioutsioukis et al., 2010; Lang et al., 2014; Lv et al., 2019; Wang et al., 2008).

The emissions related to traffic activities are responsible for the largest parcel of air pollutant concentration in urban areas (Andrade et al., 2017; D'Angiola et al., 2010). The impacts on the environment and human health are even larger in urban centers of developing countries where the regulation of vehicular emission is still a big challenge (Zhang and Batterman, 2013). Vehicular

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emission inventories provide essential information about emissions trends, main substances emitted, and major polluters. They allow the identification of the best available control technologies and provide future perspectives (Gómez et al., 2018; Gong et al., 2017; Lang et al., 2014).

In Brazil, road transport is responsible for most of the cargo handling and the fleet grows 12% a year (OICA, 2020). Vehicle ownership has increased dramatically since 1970 (DENATRAN, 2018a), going from about 24 persons per vehicle to about 4.8 in 2017. Therefore, efforts to create inventories that include the local details are scarce (Alonso et al., 2010; Gallardo et al., 2012; Huneeus et al., 2020). National and global inventories do not provide a detailed assessment required for controlling vehicular emissions at county level and refined bottom-up inventories are restricted to large urban centers (D'Avignon et al., 2010). Each county has its fleet characteristics and other features (e.g., fuel consumption and vehicle deterioration) that are not included in national (MMA, 2014, 2011a) and global databases, such as EDGAR (Crippa et al., 2020, 2018b; Madrazo et al., 2018; Puliafito et al., 2017; Van Amstel et al., 1999). Moreover, the Brazilian fleet presents particular characteristics regarding biofuel consumption, where heavy-duty vehicles run on diesel and biodiesel, and light-duty vehicles, run on ethanol, gasoline, or any mixture of both. Flex-fuel vehicle, which uses ethanol blends, has quickly gained market share since its introduction in 2003 (Mosquim and Keutenedjian Mady, 2021).

2. Methods

2.1. The BRazilian vehicular emissions Software – BRAVES

BRAVES estimates the emission from the exhaust, evaporative losses (Diurnal, Hot Soak, Running Losses, and refueling), brakes, road, tires wear, and soil resuspension. Vehicular emissions of carbon monoxide (CO), non-methane hydrocarbon (NMHC), methane (CH_4), hydrocarbons (HC), nitrogen oxides (NOx), aldehydes (RCHO), particulate material (PM), carbon dioxide (CO_2), nitrous oxide (N_2O), and sulfur dioxide (SO_2) are estimated for each fleet category. The software calculates weighted EF to account for the effect of the fleet composition in terms of category, subcategory, model-year, and fuel utilization. A similar approach was used by Maes et al. (2019) in a refined bottom-up model. The vehicles categories are segregated by light-duty, commercial-light, motorcycles, and heavy-duty vehicles. Motorcycles are divided in 4 subcategories, according to cylinder capacity (cc), while heavy duties are segregated in 8 subcategories of total gross weight.

The exhaust emissions of light-duty, commercial-light, motorcycles, and heavy-duty vehicles on BRAVES is calculated through Eq. (1):

$$E_{x,c,k} = \sum_{s=subcat._1}^{s=subcat._n} \sum_{j=fuel\ type_1}^{j=fuel\ type_n} \sum_{i=year\ model_1}^{i=year\ model_n} P_{s,j,i} \times EF_{x,c,s,j,i} \times DF_{x,c} \times VE_{s,j,i} \times FC_{k,j} \times \%V_{c,j} \tag{1}$$

where, Exc_k is the emission rate in g year^{-1} of a pollutant x and category c in a county k; $P_{s,j,i}$ is a conditional probability of a vehicle of a category c being in a subcategory s, using a fuel type j, with a year-model i; EFx,c,$_{s,j,i}$ is the emission factor in g km^{-1}; DF_{xc} is the deterioration factor, $VE_{s,j,i}$ the vehicle efficiency in km L^{-1}, $FC_{k,j}$ the fuel consumption in L year^{-1}; $\%V_{cj}$ the proportion of fuel type consumed by the category. Supplementary Material (SM 1) presents a schematic view of the exhaust emissions calculations in BRAVES.

SO_2 exhaust emission on BRAVES followed the recommendation of (CETESB, 2019). It was adopted the maximum ratio of sulfur content in gasoline (0.05 g L^{-1}), diesel S10 (0.01 g L^{-1}), and diesel S500 (0.5 g L^{-1}). Eq. (2) represent the exhaust emission of SO_2 on BRAVES:

$$ESE_{c,k} = \sum_{j=fuel\ type_1}^{j=fuel\ type_n} \sum_{i=year\ model_1}^{i=year\ model_n} P_{j,i} \times SC_{j,i} \times FC_{k,j} \times \%V_{c,j} \tag{2}$$

where, $ESE_{c,k}$ are the sulfur dioxide emissions of category c in a county k; $P_{j,i}$ the conditional probability of a vehicle from a category c using a fuel type j and year-model i; $SC_{j,i}$ the sulfur content of a category c in g km^{-1}; $FC_{k,j}$ the fuel consumption (L year^{-1}), and $\%V_{cj}$ the proportion of fuel consumed by the category.

The evaporative emissions of NMHC from light-duty and commercial-light are estimated following the recommendations of the European guide of emissions inventory (Mellios and Ntziachristos, 2019), emissions inventory from São Paulo state (CETESB, 2019), and the national emission inventory of road vehicles (MMA, 2014). The diurnal evaporative emissions occur due to temperature fluctuation during the day and night, while the automobile is parked. Hot soak occur during the first hour that the vehicle is parked after normal operation and running losses occurs while the vehicle is operating. Equations (3), (4) and (5) were implemented on BRAVES to represent the diurnal, hot soak and running losses.

$$ED_{c,k} = \sum_{j=fuel\ type_1}^{j=fuel\ type_n} \sum_{i=year\ model_1}^{i=year\ model_n} P_{j,i} \times eD \times D \times N_{c,k} \tag{3}$$

where, $ED_{c,k}$ are the diurnal evaporative emission rate of non-methane hydrocarbons in g year^{-1} for category c and county k; $P_{j,i}$ the conditional probability of a vehicle from a category c using a fuel type j and year-model i; eD the daily average emission rate from the diurnal phase in g year^{-1}; D number of days in a year and $N_{c,k}$ the total number of vehicules of a category c in a county k.

$$EHS_{c,k} = \sum_{j=fuel\ type_1}^{j=fuel\ type_n} \sum_{i=year\ model_1}^{i=year\ model_n} P_{j,i} \times eHS \times DT \times VE_{j,i} \times FC_{k,j} \times \%V_{c,j} \tag{4}$$

$$ERL_{c,k} = \sum_{j=fuel\ type_1}^{j=fuel\ type_n} \sum_{i=year\ model_1}^{i=year\ model_n} P_{j,i} \times eRL \times DT \times VE_{j,i} \times FC_{k,j} \times \%V_{c,j} \tag{5}$$

where, $EHS_{c,k}$ and $ERL_{c,k}$ are the *Hot Soak* and *Running Losses* evaporative emissions of non-methane hydrocarbons in g year^{-1} for category c and county *k; $P_{j,i}$* the conditional probability of a vehicle using a fuel type *j* and year-model*i*; *eHS* and *eRL* are the emission factors by each trip (g $trip^{-1}$); *DT* the distance travelled on each trip; $VE_{j,i}$ the vehicle efficiency (km L^{-1}); $FC_{k,j}$ the fuel consumption (L $year^{-1}$), and $\%Vc_{,j}$ the proportion of fuel consumed by the category. *eHS* and *eRL* are a function of temperature and *DT* is the distance traveled in each trip. Temperature data were obtained on the Brazilian National Meteorology Institute (INMET) website (https://clima.inmet.gov.br/NormaisClimatologicas) and DT on the Brazilian Association of Public Transportation (ANTP, 2012).

Gasoline and ethanol refueling emissions were considered on BRAVES due to its high volatility. We adopted the gasoline and ethanol volatility rate of 1.14 g L^{-1} and 0.37 g L^{-1}, similarly to CETESB (2019). Equation (6) were employed to estimate the refueling emissions:

$$ER_{c,k} = VR_j \times FC_{k,j} \times \%V_{c,j} \tag{6}$$

where, $ER_{c,k}$ is the refuelling emission of a category c in a county k, VR_j the volatility rate; $FC_{k,j}$ the fuel consumption, and $\%Vc_{,j}$ the proportion of fuel consumed by the category.

PM emissions from brake, tires, and road wear were calculated through Equation (7), following the recommendations of the European Emission Inventory Guidebook (Ntziachristos and Boulter, 2009). According to this guidebook, all wear emissions are considered as PM with an aerodynamic diameter smaller than 10 μm (PM_{10}).

$$EW_{c,k} = \sum_{j=fuel\ type_1}^{j=fuel\ type_n} \sum_{i=year\ model_1}^{i=year\ model_n} P_{j,i} \times \mathrm{EF_c} \times VE_{j,i} \times FC_{k,j} \times \%V_{c,j} \tag{7}$$

where, $EW_{c,k}$ are the wear emissions from brakes, road and tires of category c in a municipality k; $P_{j,i}$ the conditional probability of a vehicle from a category *c* using a fuel type *j* and year-model *i*; *EFc* the wear emission factor of a category c in g km^{-1}; $VE_{j,i}$ the vehicle efficiency (km L^{-1}); $FC_{k,j}$ the fuel consumption (L $year^{-1}$), and $\%Vc_{,j}$ the proportion of fuel consumed by the category.

A simplified method for estimating the soil resuspension emissions was implemented on BRAVES, as presented by Vicentini (2011) (Equation (8)). An EF value of 0.062 g km^{-1} de PM_{10} was adopted.

$$ES_{c,k} = \sum_{j=fuel\ type_1}^{j=fuel\ type_n} \sum_{i=year\ model_1}^{i=year\ model_n} P_{j,i} \times 0.062 \times VE_{j,i} \times FC_{k,j} \times \%V_{c,j} \tag{8}$$

where, $ES_{c,k}$ is the soil resuspension emission of a category *c* in county k; $P_{j,i}$ the conditional probability of a vehicle *c* using a fuel type *j* and year-model *i*; $VE_{j,i}$ the vehicle efficiency (km L^{-1}); $FC_{k,j}$ the fuel consumption (L $year^{-1}$), and $\%Vc_{,j}$ the proportion of fuel consumed by the category.

According to the EEA emission inventory guidebook (Ntziachristos and Zissis, 2019), PM exhaust emission mainly falls in $PM_{2.5}$ size. Therefore, all PM from exhaust emissions is considered as $PM_{2.5}$ in BRAVES. Since the non-exhaust emissions (soil resuspension, tire, road, and brake wear) are mostly in the PM_{10} range (Ntziachristos and Boulter, 2009), these emissions are all considered as PM_{10}.

BRAVES considers that all fuel sold in a county is consumed inside the same county. Also, a fleet of a county runs exclusively in the same county. All codes and scripts were written in MATLAB®. SM 2 contains all codes and a readme file to provide information for the users who may want to run BRAVES. BRAVES is registered at the Brazilian National Institute of Industrial Property under the number 512020001978-1. The code is also freely available at the personal website of the second author (https://hoinaski.prof.ufsc.br/BRAVES/). On this website, the users could find upgrades, inputs, and outputs from BRAVES.

2.2. BRAVES inputs

The EFs were obtained from samplings conducted by the Environmental Company of São Paulo State (CETESB, 2019), which provides emission factors of new vehicles for the categories light-duty, commercial light, motorcycles and heavy-duty (https://cetesb.sp.gov.br/veicular/relatorios-e-publicacoes/). According to National Registry of Automobile Vehicles (RENAVAM) elaborated by the

Table 1
Definition of vehicle categories used in this work.

Category	Motorization/ Fuel		Class	Definition
Light – Duty	Otto	Gasoline C Hydrous ethanol *Flex Fuel*	Automobiles, others;	Self-propelled vehicle for the transport of passengers, with a capacity for up to eight people, excluding the driver.
Commercial Light	Otto	Gasoline C Hydrous ethanol *Flex Fuel*	Pickup truck, Utility vehicle;	Vehicle designed for the transport of people or cargo, with a total gross weight of up to 3.856 kg.
	Diesel			
Motorcycles	Otto	Gasoline C Hydrous ethanol	Moped, Motorcycle, scooter, Quadricycle, tricycle;	Two- or three-wheel motor vehicle, driven by a driver in a seated or mounted position.
Heavy-Duty	Diesel		Truck, Truck Tractor, Platform Chassis, Crawler Tractor, Wheeled Tractor, Bus and Minibus	Motor vehicle for the transport of cargo, with bodywork, and a total gross weight exceeding 3.856 kg. Public transport vehicle.

National Department of Traffic (DENATRAN, 2018a), the Brazilian vehicular fleet is composed by 21 categories and e 18 fuel/motorization types, which was grouped to match the corresponding classification adopted by (CETESB, 2019) (Table 1). Some RENAVAM categories were exclude for not being considered as potential emitters (tram, trailer, semi-trailer, and sidecar).

In CETESB categorization, light-duty and commercial-light categories are not segregated in subcategories, while motorcycles and heavy-duty vehicles are. Motorcycles EF are subcategorized by cylinder capacity (cc). Until 2009, motorcycles subcategories were: (a) equal or below to 150 cc, (b) from 151 to 500 cc, and (c) above 501 cc. From 2010 onwards, subcategories are: (a) equal or below to 150 cc, and (b) above 150 cc. The number of motorcycles in each subcategory were obtained on a report from the Brazilian Association for Motorcycles and Similar Manufacturers (ABRACICLO, 2018) and RENAVAM (https://www.gov.br/infraestrutura/pt-br/assuntos/transito/conteudo-denatran/estatisticas-frota-de-veiculos-denatran).

Heavy-duty vehicles are categorized by trucks and buses. Trucks are subcategorized according to their total gross weight: (a) demi-light, (b) light, (c) average, (d) demi-heavy, and (e) heavy. Buses are subcategorized as transit bus, minibus, and express bus. The number of heavy-duty vehicles and their subcategory definitions were obtained from the National Association for Automobile Vehicles (ANFAVEA, 2018) and DENATRAN (2018).

The annual volume of fuel consumed in each county was provided by the Brazilian National Agency for Oil, Natural Gas and Biofuel (ANP) (https://www.gov.br/anp/pt-br/centrais-de-conteudo/dados-abertos/vendas-de-derivados-de-petroleo-e-biocombustiveis). A fraction of fuel consumed by road transportation was obtained on National Energy Balance (BEN) (https://www.epe.gov.br/pt/publicacoes-dados-abertos/publicacoes/balanco-energetico-nacional-ben). A table in SM 3 presents the motorization and fuel type of each category and year-model. Data from 2013 to 2018 were used in this work.

The proportion of fuel used by each category followed the national emission inventory of road vehicles (MMA, 2014). A table in SM 4 summarizes the percentage of gasoline and ethanol consumed by flex-fuel vehicles. In counties where no consumption of ethanol was reported, we considered that flex-fuel vehicles were fueled with gasoline.

A Deterioration Factor (DF) is included on EF due to vehicle-use intensity. We applied the DF each 5 year of use, which represent and average of 80.000 km of mileage (ANTP, 2012). The increment on FE due to DF is summarized in SM 5. We have adopted the same criteria as other reference studies for the Deterioration Factor in Brazil (CETESB, 2019; MMA, 2011). An analysis conducted by MMA (2011) using 200 testing cases between 2003 and 2007 showed that increment in emissions with mileage was not significant. MMA (2011) only observed an increment in the emission factors above 80.000 km of mileage.

Vehicle scrappage was also considered, according to the national emission inventory of road vehicles (MMA, 2014, 2011a). For motorcycles scrappage, we followed the recommendations of the Rio de Janeiro state emissions inventory (Souza et al., 2013). The maximum lifetime of a vehicle was limited to 40 years. The curve and associated equations for each vehicle category to calculate the scrappage is presented in SM 6 and SM 7.

The emission factors from CETESB led to the underestimation of the evaporative emissions in BRAVES. It is also observed in VEIN (Ibarra-Espinosa et al., 2020). BRAVES also estimates evaporative emissions using emissions factors from the European Council emissions guidelines provided by the European Environmental Agency (EEA) (Mellios and Ntziachristos, 2019). We use the emission factor from Tier 2 of EEA, which is compatible with BRAVES. Since CETESB does not provide the emission factors of evaporative emissions from motorcycles, BRAVES do not estimate evaporative emissions of this category. A comparison between the CETESB and EEA evaporative emission factors for the light and light commercial categories is presented in SM 8. The European Guide presents a

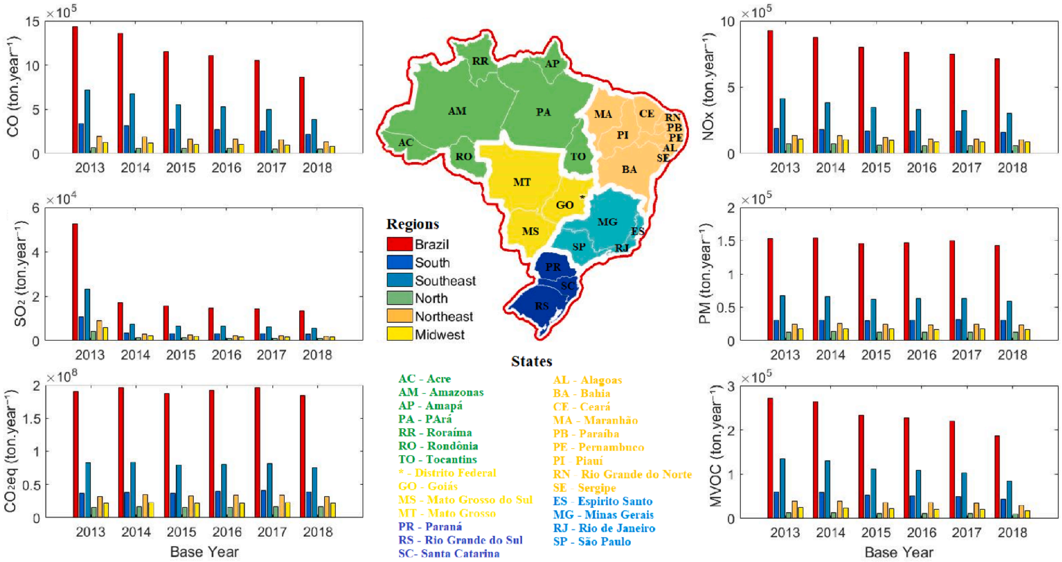

Fig. 1. Vehicular emissions of CO, SO_2, CO_2eq, NOx, PM, and NMVOC in Brazil from 2013 and 2018 estimated by BRAVES using emission factors from CETESB.

higher EF for the Diurnal process, while CETESB tends to overestimate the emissions of the Hot Soak and Running Losses processes.

3. Results and discussion

3.1. Vehicular emissions in Brazil using BRAVES

The total vehicular emissions estimated by BRAVES in 2018 are 861,980 tons of CO, 712,261 tons of NOx, 13,516 tons of SO_2, 142,961 tons of PM, 183.9x10^6 tons of CO_2eq, and 186,613 tons of NMVOC (Fig. 1). Emissions of CO, NOx, SO_2 e CO_2eq are exclusively from the exhaust, while PM is emitted by exhaust (14,8%), tires and brakes wear (53.9%), road wear (5.4%), and soil resuspension (25.9%) (Fig. 2). NMVOC is originated from exhaust (56.0%), permeation, diurnal, and hot soak evaporation process (16.9%), and refueling (27.1%) (Fig. 3).

Except for CO_2eq and PM, all vehicular emissions have decreased since 2013 (Fig. 1). SO_2 emissions dropped dramatically, especially from 2013 to 2014, when the sulfur content of gasoline reduced to lower levels (from greater than 200 ppm before 2013 to 50 ppm onwards) (PETROBRAS, 2019). Sulfur content in diesel also dropped from 1800 ppm to small than 500 ppm (CETESB, 2019). From 2013 to 2018, a small reduction of PM emissions is observed since it is mostly emitted by soil resuspension, tires, brakes, and road wear, which are hard to control.

CO and NOx are regulated by the policy programs for controlling vehicular emissions and effectively controlled by improving vehicle efficiency and installing catalytic converters (Gerard and Lave, 2005; Hao et al., 2006; Pacheco et al., 2017). Most of CO emission is avoided by optimizing the excess air and increasing the amount of oxygen to the combustion, which increase the CO_2 in the other hand (Ji and Wang, 2009). While the catalytic converters efficiency for CO_2 removal is low, this pollutant is hardly removed from exhaust emissions (Becker et al., 2000; Dasch, 1992; Lipman and Delucchi, 2002). Therefore, fuel consumption drives the CO_2eq interannual variability (Fig. 1).

Analyzing the spatial variability, the southeast region is the major source of vehicular emissions for all pollutants in Brazil. This region is the most urbanized and owns the largest fleet and fuel consumption. Southern Brazil is the second-largest emitter, followed by northeast, midwest, and north regions.

It can be observed in Fig. 2 that emissions of soil resuspension, road, tires, and brake wear showed practically no difference among the years, even though the fleet has increased in Brazil (DENATRAN, 2018b; Mosquim and Keutenedjian Mady, 2021). In BRAVES, the emissions are a function of emission factors and fuel consumption. The emission factor incorporates the fleet characteristics, consequently, the emission control techniques of each county's fleet. Fuel consumption represents the activity rate in BRAVES. Since the activity rate fluctuates over the years and the fleet technology has changed since 2013, we do not observe a significant difference for some emissions among the years. In these cases, we could see a counterbalance between the increase/decrease in fuel consumption and emission factors.

Vehicle exhaust NMVOC emissions reduced significantly over the years (Fig. 3), reflecting on the interannual variability of this

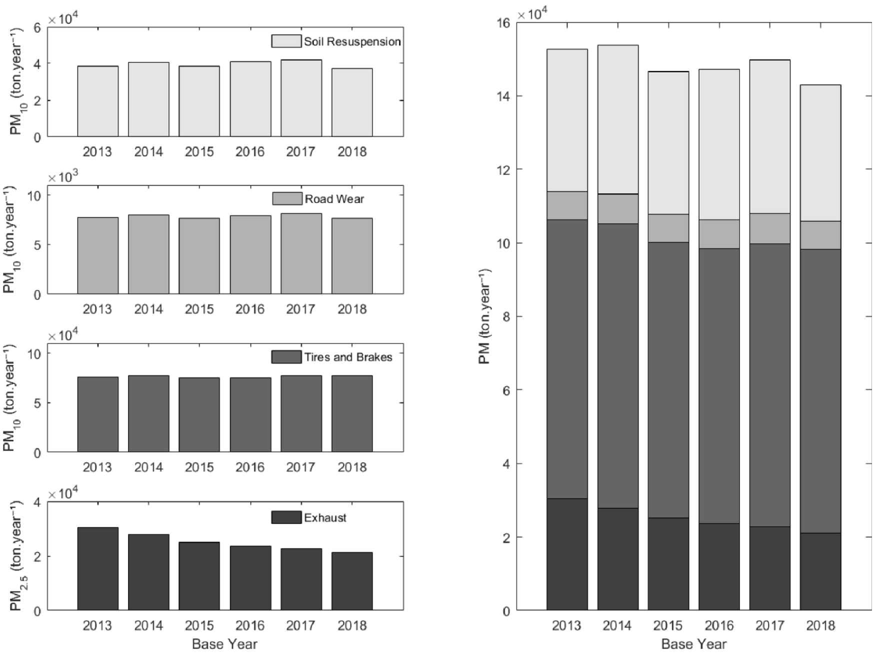

Fig. 2. Vehicular emissions of PM in Brazil from 2013 and 2018 by formation process (soil resuspension, road wear, tires and brakes, and exhaust) estimated by BRAVES using emission factors from CETESB. Emissions of PM_{10} from soil resuspension, tires, road, brake wear. Emission of $PM_{2.5}$ from exhaust.

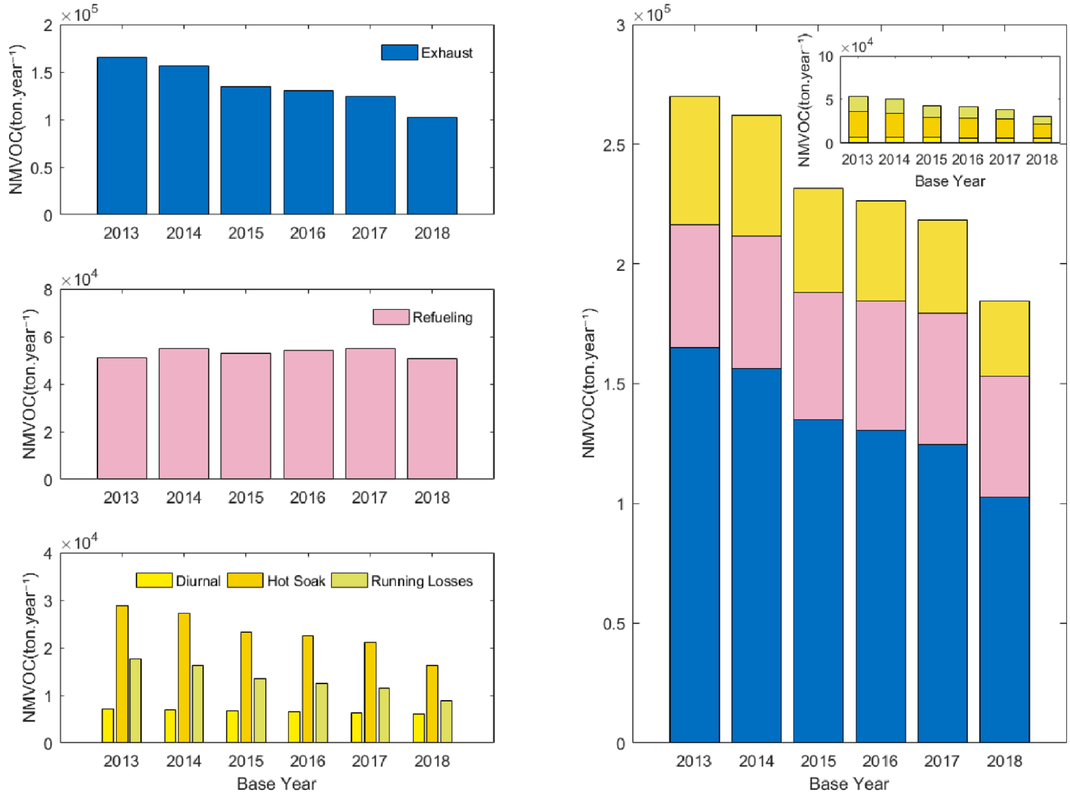

Fig. 3. Evaporative emissions of NMVOC in Brazil from 2013 and 2018 by formation process (exhaust, refueling, diurnal, hot soak, running losses) estimated by BRAVES using emission factors from CETESB.

pollutant. Improving combustion efficiency and emission controlling techniques in new vehicles reduced the emission of pollutants including NMVOC, suggesting that the program for controlling vehicular emissions in Brazil has obtained positive effects (CETESB, 2019; Policarpo et al., 2018; Souza et al., 2013; Szwarcfiter et al., 2005). On the other hand, evaporative NMVOC emissions remained at the same levels from 2013 to 2018 (Fig. 3), increasing their proportion to the total NMVOC vehicular emissions year by year. The evaporative process is still not being regulated by the program for controlling vehicular emissions in Brazil (Andrade et al., 2017).

BRAVES also provides estimates of evaporative emissions using the emission factors from EEA. Figure SM 9 shows a comparison between BRAVES estimates using emission factors from CETESB and EEA. The analysis confirms that BRAVES underestimate the evaporative emissions when using emission factors from CETESB. Therefore, CETESB emission factors for evaporative NMVOC are low (Ibarra-Espinosa et al., 2020). Even though evaporative emission factors from EEA seem to be more accurate than CETESB, it may be significantly underestimated for the Brazilian reality, since it does not account for the vintage of the vehicle fleet, deterioration, fuel vapor pressure, and temperatures. It is demonstrated by Koupal and Palacios (2019) that evaporative NMVOC emissions are influenced by the ethanol content, which is also high in Brazil. In Brazil, the limit for ethanol blends ranges from 25 to 27% in 2021 (BRASIL, 2015). According to Ibarra-Espinosa et al. (2020), the only evaporative emission factors available in the literature with Brazilian chemical composition in the fuel comes from CETESB. Future studies could include the influence of vehicle fleet, deterioration, fuel vapor pressure, and temperatures in the evaporative emission factors and their influence on BRAVES estimates.

Recent updates to Mexico's gasoline specification allow ethanol blends up to 10 percent (E10) (Koupal and Palacios, 2019). Using a version of MOVES adapted to Mexico which incorporates research on the effect of gasoline fuel properties on exhaust and evaporative vehicle emissions, Koupal and Palacios (2019) demonstrated an increase in VOC emissions from 13 to 18 percent by the introduction of E10.

Figure SM 10 presents the comparison of BRAVES weighted emission factors for CO, NOx, $MP_{2,5}$, NMVOC and N_2O exhaust, using the CETESB and EEA emission factors. The European guide emission factors are highest for the light vehicle, light commercial and motorcycle categories. However, CETESB emission factors are higher for the heavy vehicle category.

Comparing the emissions by fleet category, light-duties emits the most of vehicular CO and NMVOC, while heavy-duties contribute to the largest proportion of NOx, SO_2, PM, and CO_2eq. The same pattern is observed all over the Brazilian regions (Fig. 4). This result reveals that heavy duties would be an effective target for controlling NOx, SO_2, PM, and CO_2eq emissions, since it accounts for only 4.3% of the Brazilian overall fleet (DENATRAN, 2018). Motorcycles are great emitters of CO and NMVOC, representing 27.1% of the Brazilian fleet. Commercial lights are the smallest contributors to the total vehicular emissions. SM 11 shows the proportion of licensed vehicles and fuel consumption in Brazilian regions in 2018.

São Paulo is the Brazilian state with the largest vehicular emission for all pollutants estimated by BRAVES in 2018 (Fig. 5), despite relative success in reducing vehicular emissions since 2013. Great spatial variability of vehicular emissions is observed among the Brazilian states (boxplots in SM 12). Except for CO_2eq and PM, all states have been reducing their vehicular emissions. CO_2eq emissions have increased in 12 states from south, midwest, and north. NOx presents the largest variability of reduction/increase among the Brazilian states over the years. The vehicular emissions by counties and its spatial variability estimated by BRAVES are presented in SM13 and SM14.

3.2. Comparing BRAVES to global, national and local inventories

BRAVES were compared to global, national, and local vehicular emission inventories, under national, state, and county scales. On a country scale, BRAVES estimates were much lower than those from Emissions Database for Global Atmospheric Research (EDGAR v4.3.2 and EDGAR v5.0) (https://edgar.jrc.ec.europa.eu/) (Crippa et al., 2020, 2018a). The Fractional Bias between BRAVES and vehicular emissions from EDGARv5.0 was −146% for CO, −143% for NMVOC, −73% for $PM_{2,5}$, −134% for PM_{10}, −59% for NOx, and −21% for CO_2eq (Table 2). As reported previously by Madrazo et al. (2018), most of the road transport EFs are overestimated in EDGAR. Large discrepancies were also found by Huneeus et al. (2020) between EDGAR and local/national city emissions data for the same domain. Huneeus et al. (2020) do not recommend using a global emission inventory to derive city emissions.

Differences between BRAVES and the national inventory (MMA, 2014) are relatively low in national scale. BRAVES estimates are

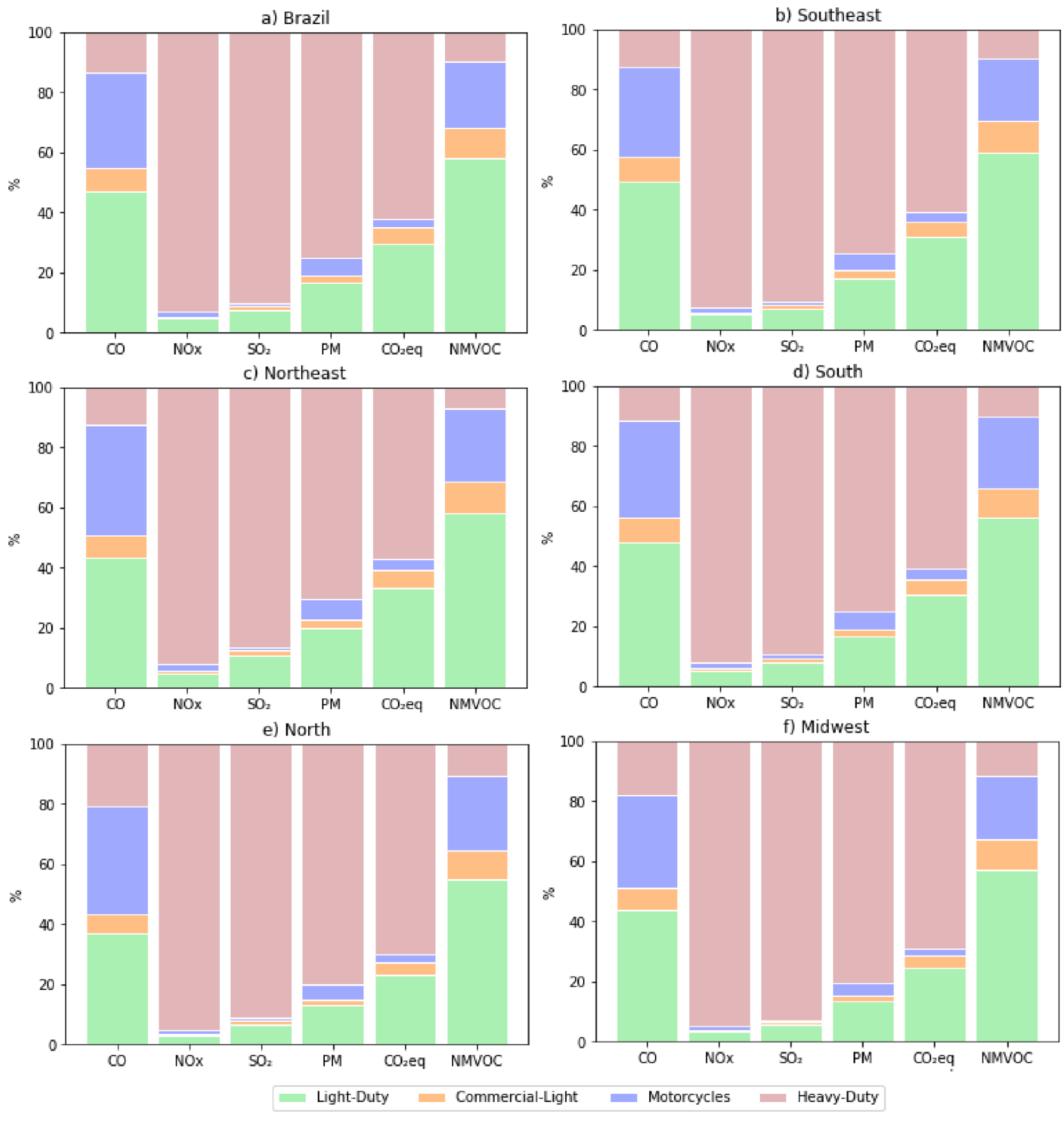

Fig. 4. Contribution of fleet categories to the total vehicular emissions of CO, NOx, SO_2, PM, CO_2eq, and NMVOC in Brazil using emission factors from CETESB.

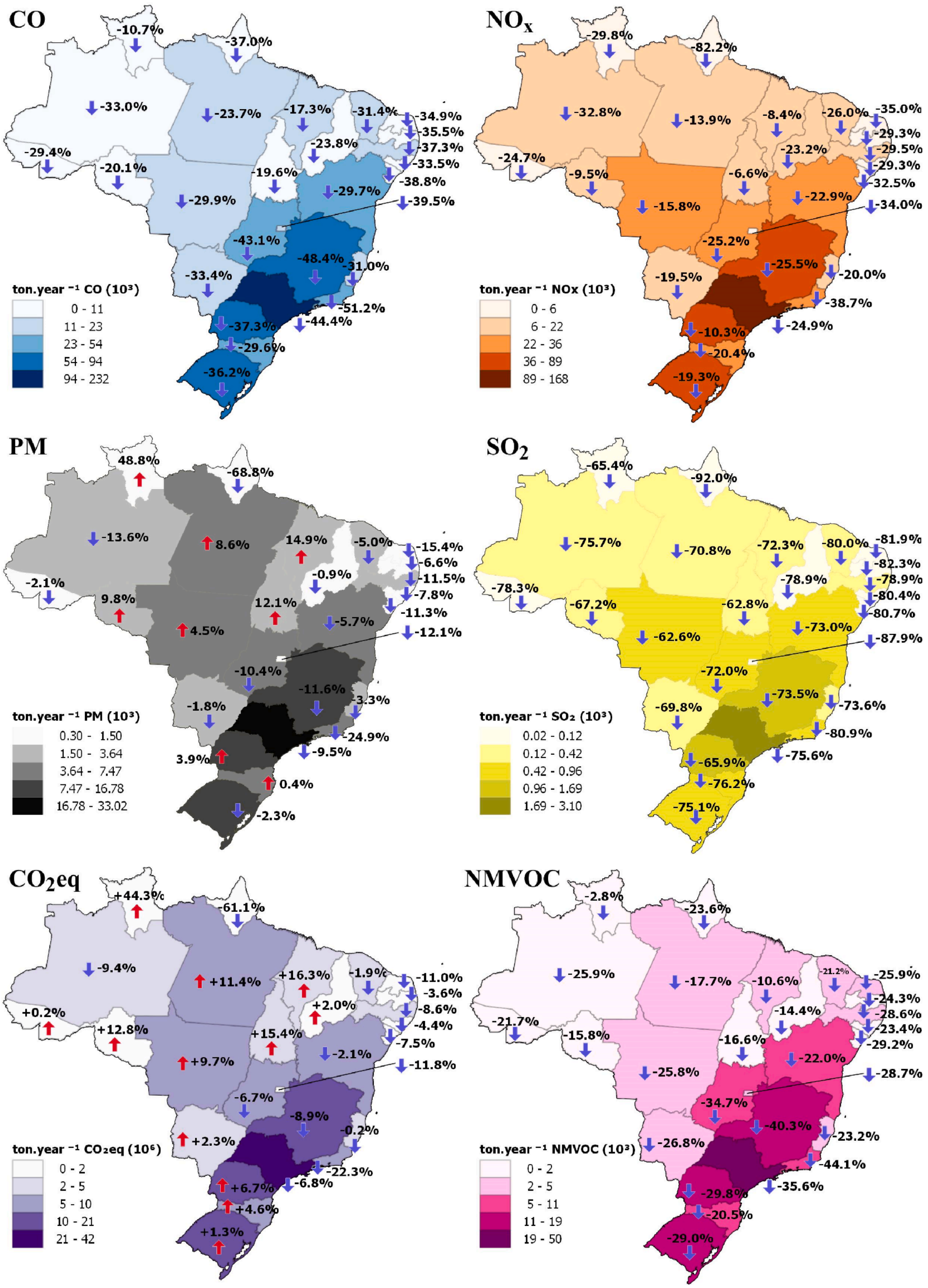

Fig. 5. Vehicular emissions in Brazilian states estimated by BRAVES in 2018 using emission factors from CETESB. Reduction/increase percentage from 2013 to 2018 indicated by arrows.

slightly higher for CO (14%) and NMVOC (9%), and lower for CO_2eq, (-11%), NOx (-17%), and $PM_{2.5}$ (-20%) (Table 2). The highest discrepancy was observed for the PM_{10} (130%). The advantage of using BRAVES relies on the ability to estimate at county level and to address local fleet and fuel consumption characteristics.

In state scale, BRAVES were compared to the vehicular emissions inventory from the state of São Paulo (CETESB, 2014) and Minas

Table 2
Bias and fractional bias between BRAVES estimates and other vehicular inventories in different spatial scales. BRAVES estimates using emission factors from CETESB.

Reference	Spatial Scale	Year	BIAS (Fractional Bias)					
			CO ton. year^{-1}	NOx ton. year^{-1}	$PM_{2.5}$ ton. year^{-1}	PM_{10} ton. year^{-1}	CO_2eq 10^6 ton.year^{-1}	NMVOC ton. year^{-1}
	Metropolitan Region Scale							
SEEG v8.0 (De Azevedo et al., 2018)	São Paulo/SP (39 counties)	2014	−25,862 (0.16)	14,621 (−0.22)	–	–	0.14 (−0.01)	−8,020 (0.28)
(CETESB, 2015)			−8,031 (0.05)	−4,194 (0.07)	−420 (0.25)	–	−1.82 (0.12)	1,751 (−0.05)
(Ibarra-Espinosa et al., 2020)			6,479 (−0.04)	15,026 (−0.23)	377 (−0.18)	–	4.95 (−0.27)	926 (−0.03)
SEEG v8.0 (De Azevedo et al., 2018)	Campinas/SP (20 counties)	2014	−1,195 (0.04)	4,938 (−0.27)	–	–	0.00 (0.00)	−876 (0.15)
(CETESB, 2015)			3,080 (−0.09)	−1,796 (0.12)	−127 (0.29)	–	−0.50 (0.14)	914 (−0.14)
(Ibarra-Espinosa et al., 2020)			66,263 (−1.02)	36,179 (−1.07)	1,993 (−1.33)	–	5.34 (−0.83)	24,605 (−1.33)
SEEG v8.0 (De Azevedo et al., 2018)	Vale do Paraíba e Litoral Norte/SP (39 counties)	2014	−4,305 (0.19)	2,746 (−0.21)	–	–	0.05 (−0.02)	−1,149 (0.27)
(CETESB, 2015)			1,986 (−0.08)	−1,691 (0.15)	−129 (0.38)	–	−0.34 (0.13)	491 (−0.10)
(Ibarra-Espinosa et al., 2020)			628 (−0.02)	−146 (0.01)	133 (−0.29)	–	−0.61 (0.25)	3,766 (−0.56)
SEEG v8.0 (De Azevedo et al., 2018)	Sorocaba/SP (27 counties)	2014	−1,299 (0.07)	2,606 (−0.25)	–	–	0.00 (0.00)	426 (−0.15)
(CETESB, 2015)			1,310 (−0.07)	−438 (0.05)	−51 (0.18)	–	−0.27 (0.13)	1,383 (−0.41)
(Ibarra-Espinosa et al., 2020)			5,205 (−0.24)	2,500 (−0.24)	266 (−0.60)	–	−0.01 (0.00)	5,082 (−0.97)
SEEG v8.0 (De Azevedo et al., 2018)	Baixada Santista/SP (9 counties)	2014	1,583 (−0.16)	2,260 (−0.30)	–	–	−0.01 (0.01)	−49 (0.03)
(CETESB, 2015)			4,208 (−0.37)	−320 (0.05)	−12 (0.06)	–	−0.17 (0.12)	692 (−0.30)
(Ibarra-Espinosa et al., 2020)			15,525 (−0.91)	9,247 (−0.83)	601 (−1.22)	–	0.88 (−0.44)	6,202 (−1.23)
SEEG v8.0 (De Azevedo et al., 2018)	Rio de Janeiro/RJ (21 counties)	2013	−10,946 (0.18)	9,829 (−0.26)	–	–	0.33 (−0.05)	−2,861 (0.26)
(INEA, 2016) - A			−25,256 (0.47)	9,793 (−0.26)	−225 (0.22)	−2,109 (0,62)	–	−4,974 (0.50)
(INEA, 2016) - B			−40,802 (0.89)	−17,551 (0.74)	−763 (1.04)	−3,526 (1,32)	–	−8,419 (1.03)
SEEG v8.0 (De Azevedo et al., 2018)	Grande Vitória/ES (5 counties)	2015	616 (−0.06)	530 (−0.08)	–	–	−0.20 (0.13)	−350 (0.19)
(IEMA, 2019)			1,594 (−0.15)	2,679 (−0.35)	95 (−0.35)	−629 (−0.40)	–	3,206 (−0.89)
SEEG v8.0 (De Azevedo et al., 2018)	Florianópolis/SC (5 counties)	2017	1,068 (−0.14)	941 (−0.41)	–	–	0.00 (0.00)	−357 (0.25)
(Maes et al., 2019)			2,302 (−0.28)	1,188 (−0.49)	47 (−0.64)	–	–	–
SEEG v8.0 (De Azevedo et al., 2018)	Fortaleza/CE (6 counties)	2015	3,299 (−0.23)	3,191 (−0.45)	–	–	0.10 (−0.05)	−199 (0.07)
(Policarpo et al., 2018)			−899 (0.07)	−2,700 (0.66)	−41 (0.32)	–	–	−1,298 (0.60)
	State Scale							
SEEG v8.0 (De Azevedo et al., 2018)	State of São Paulo (645 counties)	2013	−42,227 (0.11)	52,783 (−0.21)	–	–	0.42 (−0.01)	−12.441 (0.17)
(CETESB, 2014)						–	−4.94 (0.11)	

(continued on next page)

Table 2 (*continued*)

Reference	Spatial Scale	Year	BIAS (Fractional Bias)					
			CO ton. $year^{-1}$	NOx ton. $year^{-1}$	$PM_{2,5}$ ton. $year^{-1}$	PM_{10} ton. $year^{-1}$	CO_2eq 10^6 $ton.year^{-1}$	NMVOC ton. $year^{-1}$
			5,120 (−0.01)	–32,043 (0.15)	−2,308 (0.35)			−6.870 (0.09)
SEEG v8.0 (De Azevedo et al., 2018)	State of Minas Gerais (853 counties)	2015	−43,975 (0.29)	8,358 (−0.07)	–	–	−0.12 (0.01)	−10.015 (0.36)
(Santos et al., 2021)			−72,865 (0.52)	−2,576 (0.02)	78 (−0.02)	−10,531 (1,20)	–	−6.653 (0.23)
	National Scale							
SEEG v8.0 (De Azevedo et al., 2018)	Country BRAZIL (5570 counties)	2013	−96,091 (0.07)	271,550 (−0.26)	–	–	5.07 (−0.03)	−36,322 (0.14)
(MMA, 2014)		2012	−181,903 (0.14)	176,049 (−0.17)	6,909 (−0.20)	−96,420 (1,30)	22.84 (−0.11)	–22,430 (0.09)
EDGARv4.3.2 (Crippa et al., 2018a)		2012	7,557,998 (−1.45)	634,542 (−0.51)	35,007 (−0.73)	–	33.89 (−0.16)	1,350,215 (−1.42)
EDGARv5.0 (Crippa et al., 2020)		2013	7,695,668 (−1.46)	766,519 (−0.59)	34,685 (−0,73)	−98,104 (1.34)	43.74 (−0.21)	1,377,575 (−1.43)

Gerais state (Santos et al., 2021). In São Paulo, the differences between BRAVES and the bottom-up model used by CETESB (2014) range from –1% to 35%, while in Minas Gerais range from −2% to 52%. If we compare PM_{10} emissions in BRAVES and the Minas Gerais state inventory, the Fractional Bias is 120%. The inventory of the state of São Paulo does not estimate particulate matter emissions from non-exhaustive process.

In local scale, BRAVES reached a good agreement for most of the cases, except for Campinas, Sorocaba, and Baixada Santista in a study conducted by Espinosa et al. (2020), using the software Vehicular Emission Inventory (VEIN). VEIN is a high spatial and temporal resolution model which accounts for the classification of vehicles into several categories, different options of emission factors and specification of pollutants, and input traffic from traffic simulations or other network-based sources. Relatively small bias between BRAVES and VEIN was observed in São Paulo and Vale do Paraiba.

Even though BRAVES uses a different approach to calculate the emission factors, the comparison to independent inventories in estimating CO_2eq shows that BRAVES agrees with VEIN, which is a very refined method (Table 2). The fractional bias between these models ranges from −83% to 25%, where BRAVES underestimate the VEIN's emissions (Table 2). Future work would clarify the temporal e spatial differences between BRAVES and other independent measurements.

Vehicle emissions in the metropolitan region of Rio de Janeiro were estimated by INEA (2016) using a top-down (A) and bottom-up (B) approach. BRAVES has estimated higher values when compared to INEA (2016) top-down approach, with the exception of NOx emissions. In general, the fractional bias is relatively small, especially for estimates with a top-down approach.

Comparing to the very refined emission inventory in Great Vitoria – ES (IEMA, 2019), BRAVES reached a relatively good agreement for CO, NOx, $PM_{2,5}$ and PM_{10} emissions (Table 2). However, the model underestimated the Non-Methane Volatile Organic Compounds (NMVOC) in −89%. The emissions inventory developed in Vitoria used several on-site measurements, including near-road monitoring. BRAVES uses a fixed emission factor for estimating PM emissions, being insensitive to road type (paved or unpaved), silt loadings, and detailed local conditions. This drawback may lead to errors in estimating PM emissions which could explain the bias when compared to Vitoria's inventory. Further studies could explore the differences between outputs from BRAVES and sophisticated models and inventories.

BRAVES was also compared to high-resolution inventories in Florianópolis (Maes et al., 2019) and Fortaleza (Policarpo et al., 2018), reaching good agreement for most of the estimated pollutants (Table 2). In Florianópolis, Maes et al. (2019) do not considered wear and soil resuspension emissions.

The Greenhouse Gas Emission and Removal Estimating System (SEEG) (De Azevedo et al., 2018) is an emission inventory system that estimates the greenhouse gas (GHG) emissions on each Brazilian county. The SEEG presents the emissions by sectors, such as energy (fuel consumption), agriculture, land use change, industrial process, and waste management. Comparing BRAVES and SEEG (energy sector), the models presented close estimates of CO and CO_2eq on national, state, and county scales. A larger bias is observed in NOx and NMVOC emissions. A strong correlation between BRAVES and SEEG estimates is observed in the state of São Paulo (SM 15). BRAVES presents a more detailed characterization of vehicular emissions, including emissions of each fleet category. PM emissions are calculated for each formation process in BRAVES, while in SEEG this pollutant is not estimated. BRAVES and SEEG have different design purposes. BRAVES is a vehicular emission inventory system focused on air quality issues and criteria pollutants, and SEEG is a GHG inventory.

A full dataset with outputs from BRAVES from 2013 to 2018 can be found on SM16 (total emissions by county) and SM17 (emissions by category and county). The weighted emission factors estimated by BRAVES for each vehicle category and county are presented in SM 18.

We do not evaluate the uncertainty associated with the inputs and parameters in BRAVES, which could reflect the accuracy of the results. Even though most of the methods compared in Table 2 use a similar approach for the deterioration factor, a comprehensive comparison among the methods would be required to account for the uncertainty associated with the emission factors.

4. Conclusions

In this work, we present the BRAVES model for estimating vehicular emissions of CO, NMHC, CH_4, HC, NOx, RCHO, PM, CO_2, N_2O, and SO_2 from the exhaust, evaporative process, soil resuspension, road, tire, and brake wear. The model provides emissions by each fleet category and emission process.

Our results show that most vehicular emissions are decreasing in Brazil. This indicates a possible effect of policy programs for controlling vehicular emissions in Brazil. However, controlling emissions of NMVOC, CO_2eq, and PM is a big challenge.

Compared to EDGAR global inventory, BRAVES have reached better agreement with state and local inventories. Moreover, BRAVES can estimate emissions of a larger variety of pollutants at the county scale and segregate the emissions by formation process and fleet category, a major advantage over national and state inventories. BRAVES is a compromise of more sophisticated models that provide detailed information in data-scarce areas, like small counties and could be easily adapted to other developing countries. Further studies could improve the spatial and temporal resolution of BRAVES using disaggregation methods. Also, the effect of the vehicular fleet, deterioration, fuel vapor pressure, and temperatures in the evaporative emission factors and their influence on BRAVES estimates should need investigation. A sensitivity analysis including the monotonical increase in the emission factors as a function of mileage would be required. Upgraded versions of BRAVES will provide the upper and lower values of emissions estimates.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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